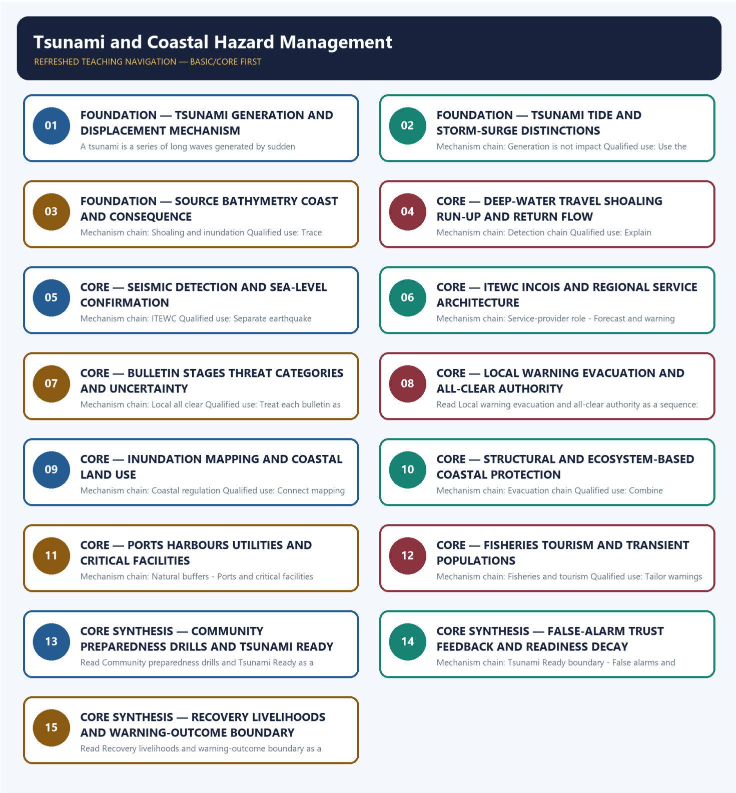
SOLVED PRACTICE WORKBOOK

Tsunami and Coastal Hazard Management - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / WORKBOOK INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Tsunami and Coastal Hazard Management - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Tsunami generation?

A. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. B. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. C. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. D. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast.

Answer: A. Explanation: A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q2. Which option preserves the risk or institutional boundary of Tsunami generation?

A. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. B. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. C. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. D. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment.

Answer: B. Explanation: A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q3. Which statement uses Tsunami generation without changing its hazard, mandate or status?

A. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. B. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. C. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. D. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment.

Answer: C. Explanation: A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q4. Which option avoids the standard UPSC close-option trap about Tsunami generation?

A. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. B. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. C. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. D. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events.

Answer: D. Explanation: A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q5. Which statement correctly identifies Tide and storm-surge boundary?

A. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. B. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. C. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. D. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast.

Answer: A. Explanation: A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q6. Which option preserves the risk or institutional boundary of Tide and storm-surge boundary?

A. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. B. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. C. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. D. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities.

Answer: B. Explanation: A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q7. Which statement uses Tide and storm-surge boundary without changing its hazard, mandate or status?

A. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. B. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. C. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. D. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities.

Answer: C. Explanation: A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with

sudden water displacement. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q8. Which option avoids the standard UPSC close-option trap about Tide and storm-surge boundary?

A. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. B. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. C. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. D. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement.

Answer: D. Explanation: A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q9. Which statement correctly identifies Generation is not impact?

A. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. B. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. C. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. D. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed.

Answer: A. Explanation: An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q10. Which option preserves the risk or institutional boundary of Generation is not impact?

A. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. B. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. C. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. D. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority.

Answer: B. Explanation: An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q11. Which statement uses Generation is not impact without changing its hazard, mandate or status?

A. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. B. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. C. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. D. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities.

Answer: C. Explanation: An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q12. Which option avoids the standard UPSC close-option trap about Generation is not impact?

A. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. B. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. C. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. D. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast.

Answer: D. Explanation: An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q13. Which statement correctly identifies Shoaling and inundation?

A. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. B. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. C. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. D. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority.

Answer: A. Explanation: Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q14. Which option preserves the risk or institutional boundary of Shoaling and inundation?

A. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. B. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. C. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. D. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities.

Answer: B. Explanation: Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q15. Which statement uses Shoaling and inundation without changing its hazard, mandate or status?

A. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. B. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. C. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. D. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities.

Answer: C. Explanation: Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q16. Which option avoids the standard UPSC close-option trap about Shoaling and inundation?

A. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. B. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. C. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. D. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed.

Answer: D. Explanation: Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q17. Which statement correctly identifies Detection chain?

A. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. B. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. C. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. D. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact.

Answer: A. Explanation: Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q18. Which option preserves the risk or institutional boundary of Detection chain?

A. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. B. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. C. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. D. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact.

Answer: B. Explanation: Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q19. Which statement uses Detection chain without changing its hazard, mandate or status?

A. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. B. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. C. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. D. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact.

Answer: C. Explanation: Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q20. Which option avoids the standard UPSC close-option trap about Detection chain?

A. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. B. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. C. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. D. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment.

Answer: D. Explanation: Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q21. Which statement correctly identifies ITEWC?

A. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. B. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. C. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. D. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact.

Answer: A. Explanation: The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q22. Which option preserves the risk or institutional boundary of ITEWC?

A. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. B. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. C. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. D. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness.

Answer: B. Explanation: The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q23. Which statement uses ITEWC without changing its hazard, mandate or status?

A. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. B. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. C. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. D. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities.

Answer: C. Explanation: The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q24. Which option avoids the standard UPSC close-option trap about ITEWC?

A. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. B. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. C. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. D. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities.

Answer: D. Explanation: The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q25. Which statement correctly identifies Service-provider role?

A. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. B. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. C. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. D. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact.

Answer: A. Explanation: ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q26. Which option preserves the risk or institutional boundary of Service-provider role?

A. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. B. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. C. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. D. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast.

Answer: B. Explanation: ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q27. Which statement uses Service-provider role without changing its hazard, mandate or status?

A. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. B. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. C. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. D. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast.

Answer: C. Explanation: ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q28. Which option avoids the standard UPSC close-option trap about Service-provider role?

A. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. B. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. C. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. D. ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority.

Answer: D. Explanation: ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q29. Which statement correctly identifies Forecast and warning?

A. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. B. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. C. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. D. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast.

Answer: A. Explanation: A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. The other options change the hazard or risk category, institutional owner,

legal instrument, warning stage, evidence rung or implementation status.

Q30. Which option preserves the risk or institutional boundary of Forecast and warning?

A. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. B. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. C. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. D. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness.

Answer: B. Explanation: A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q31. Which statement uses Forecast and warning without changing its hazard, mandate or status?

A. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. B. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. C. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. D. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast.

Answer: C. Explanation: A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q32. Which option avoids the standard UPSC close-option trap about Forecast and warning?

A. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. B. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. C. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. D. A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact.

Answer: D. Explanation: A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q33. Which statement correctly identifies Local all clear?

A. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. B. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. C. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. D. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness.

Answer: A. Explanation: ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q34. Which option preserves the risk or institutional boundary of Local all clear?

A. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. B. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. C. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. D. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills.

Answer: B. Explanation: ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q35. Which statement uses Local all clear without changing its hazard, mandate or status?

A. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. B. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. C. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. D. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk.

Answer: C. Explanation: ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q36. Which option avoids the standard UPSC close-option trap about Local all clear?

A. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. B. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. C. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. D. ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities.

Answer: D. Explanation: ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q37. Which statement correctly identifies Inundation mapping?

A. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. B. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. C. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. D. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills.

Answer: A. Explanation: Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q38. Which option preserves the risk or institutional boundary of Inundation mapping?

A. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. B. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. C. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. D. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk.

Answer: B. Explanation: Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q39. Which statement uses Inundation mapping without changing its hazard, mandate or status?

A. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. B. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. C. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. D. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure.

Answer: C. Explanation: Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q40. Which option avoids the standard UPSC close-option trap about Inundation mapping?

A. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. B. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. C. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. D. Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness.

Answer: D. Explanation: Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q41. Which statement correctly identifies Coastal regulation?

A. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. B. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. C. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. D. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure.

Answer: A. Explanation: CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q42. Which option preserves the risk or institutional boundary of Coastal regulation?

A. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. B. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. C. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. D. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific.

Answer: B. Explanation: CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q43. Which statement uses Coastal regulation without changing its hazard, mandate or status?

A. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. B. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. C. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. D. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone.

Answer: C. Explanation: CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q44. Which option avoids the standard UPSC close-option trap about Coastal regulation?

A. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. B. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. C. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. D. CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast.

Answer: D. Explanation: CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q45. Which statement correctly identifies Evacuation chain?

A. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. B. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. C. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. D. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure.

Answer: A. Explanation: Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage,

evidence rung or implementation status.

Q46. Which option preserves the risk or institutional boundary of Evacuation chain?

A. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. B. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. C. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. D. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific.

Answer: B. Explanation: Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q47. Which statement uses Evacuation chain without changing its hazard, mandate or status?

A. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. B. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. C. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. D. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country.

Answer: C. Explanation: Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q48. Which option avoids the standard UPSC close-option trap about Evacuation chain?

A. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. B. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. C. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. D. Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills.

Answer: D. Explanation: Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q49. Which statement correctly identifies Natural buffers?

A. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. B. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. C. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. D. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure.

Answer: A. Explanation: Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q50. Which option preserves the risk or institutional boundary of Natural buffers?

A. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. B. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. C. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. D. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific.

Answer: B. Explanation: Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q51. Which statement uses Natural buffers without changing its hazard, mandate or status?

A. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. B. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. C. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. D. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country.

Answer: C. Explanation: Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q52. Which option avoids the standard UPSC close-option trap about Natural buffers?

A. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. B. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. C. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. D. Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk.

Answer: D. Explanation: Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q53. Which statement correctly identifies Ports and critical facilities?

A. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. B. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. C. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. D. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone.

Answer: A. Explanation: Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q54. Which option preserves the risk or institutional boundary of Ports and critical facilities?

A. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. B. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. C. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. D. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone.

Answer: B. Explanation: Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q55. Which statement uses Ports and critical facilities without changing its hazard, mandate or status?

A. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. B. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. C. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. D. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty.

Answer: C. Explanation: Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q56. Which option avoids the standard UPSC close-option trap about Ports and critical facilities?

A. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. B. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. C. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. D. Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure.

Answer: D. Explanation: Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q57. Which statement correctly identifies Fisheries and tourism?

A. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. B. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. C. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. D. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country.

Answer: A. Explanation: Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q58. Which option preserves the risk or institutional boundary of Fisheries and tourism?

A. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. B. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. C. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. D. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty.

Answer: B. Explanation: Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q59. Which statement uses Fisheries and tourism without changing its hazard, mandate or status?

A. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. B. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. C. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. D. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence.

Answer: C. Explanation: Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q60. Which option avoids the standard UPSC close-option trap about Fisheries and tourism?

A. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. B. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. C. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. D. Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific.

Answer: D. Explanation: Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q61. Which statement correctly identifies Community preparedness?

A. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. B. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. C. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. D. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty.

Answer: A. Explanation: Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q62. Which option preserves the risk or institutional boundary of Community preparedness?

A. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. B. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. C. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. D. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development.

Answer: B. Explanation: Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q63. Which statement uses Community preparedness without changing its hazard, mandate or status?

A. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. B. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. C. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. D. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events.

Answer: C. Explanation: Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q64. Which option avoids the standard UPSC close-option trap about Community preparedness?

A. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. B. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. C. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. D. Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone.

Answer: D. Explanation: Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q65. Which statement correctly identifies Tsunami Ready boundary?

A. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. B. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. C. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. D. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development.

Answer: A. Explanation: UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q66. Which option preserves the risk or institutional boundary of Tsunami Ready boundary?

A. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. B. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. C. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. D. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development.

Answer: B. Explanation: UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q67. Which statement uses Tsunami Ready boundary without changing its hazard, mandate or status?

A. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. B. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. C. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. D. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events.

Answer: C. Explanation: UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q68. Which option avoids the standard UPSC close-option trap about Tsunami Ready boundary?

A. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. B. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. C. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. D. UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country.

Answer: D. Explanation: UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q69. Which statement correctly identifies False alarms and uncertainty?

A. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. B. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. C. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. D. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence.

Answer: A. Explanation: Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q70. Which option preserves the risk or institutional boundary of False alarms and uncertainty?

A. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. B. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. C. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. D. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence.

Answer: B. Explanation: Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q71. Which statement uses False alarms and uncertainty without changing its hazard, mandate or status?

A. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. B. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. C. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. D. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast.

Answer: C. Explanation: Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q72. Which option avoids the standard UPSC close-option trap about False alarms and uncertainty?

A. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. B. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. C. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. D. Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty.

Answer: D. Explanation: Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q73. Which statement correctly identifies Recovery and livelihoods?

A. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. B. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. C. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. D. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events.

Answer: A. Explanation: Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q74. Which option preserves the risk or institutional boundary of Recovery and livelihoods?

A. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. B. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. C. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. D. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events.

Answer: B. Explanation: Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q75. Which statement uses Recovery and livelihoods without changing its hazard, mandate or status?

A. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. B. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. C. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. D. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement.

Answer: C. Explanation: Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q76. Which option avoids the standard UPSC close-option trap about Recovery and livelihoods?

A. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. B. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. C. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. D. Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development.

Answer: D. Explanation: Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q77. Which statement correctly identifies Warning-outcome firewall?

A. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. B. A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. C. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. D. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement.

Answer: A. Explanation: A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q78. Which option preserves the risk or institutional boundary of Warning-outcome firewall?

A. A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. B. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. C. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. D. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed.

Answer: B. Explanation: A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q79. Which statement uses Warning-outcome firewall without changing its hazard, mandate or status?

A. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. B. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. C. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. D. An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast.

Answer: C. Explanation: A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q80. Which option avoids the standard UPSC close-option trap about Warning-outcome firewall?

A. Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. B. Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. C. The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. D. A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence.

Answer: D. Explanation: A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

The 2025 GS-I formation card and 2023 GS-I coastline card are verified cross-paper routes. The 2024 resilience card is a conservative application and does not displace Topic 01 ownership.

PYQ DEMAND CARD 1 - 2025 GS-I

Demand: Explain tsunami formation and consequences.

Status: Verified cross-paper route: Explain · 10 marks · 150 words; Geography retains physical-process ownership and this card supplies the disaster-management application.

Model solution: Tsunami generation: A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. **Tide and storm-surge boundary:** A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. **Generation is not impact:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Shoaling and inundation:** Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. **Detection chain:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Local all clear:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Evacuation chain:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Natural buffers:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on ‘PYQ DEMAND CARD 1 - 2025 GS-I', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Tsunami generation: A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. **Tide and storm-surge boundary:** A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. **Generation is not impact:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Shoaling and inundation:** Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. **Detection chain:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Local all clear:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Evacuation chain:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Natural buffers:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim:** Demand: Explain tsunami formation and consequences. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:**

Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

2. **Claim:** Status: Verified cross-paper route: Explain · 10 marks · 150 words; Geography retains physical-process ownership and this card supplies the disaster-management application. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Tsunami generation: A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. **Tide and storm-surge boundary:** A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. **Generation is not impact:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Shoaling and inundation:** Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. **Detection chain:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Local all clear:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Evacuation chain:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Natural buffers:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘PYQ DEMAND CARD 1 - 2025 GS-I', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

PYQ DEMAND CARD 2 - 2023 GS-I

Demand: Comment on the resource potential of India's coastline and highlight hazard preparedness.

Status: Verified cross-cutting route: Comment and highlight · 15 marks · 250 words; this card addresses only tsunami and coastal-preparedness dimensions.

Model solution: ITEWC: The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Inundation mapping:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Coastal regulation:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. **Evacuation chain:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Natural buffers:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Ports and critical facilities:** Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. **Fisheries and tourism:** Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. **Community preparedness:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Recovery and livelihoods:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on 'PYQ DEMAND CARD 2 - 2023 GS-I', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: ITEWC: The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Inundation mapping:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Coastal regulation:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. **Evacuation chain:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Natural buffers:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Ports and critical facilities:** Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. **Fisheries and tourism:** Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. **Community preparedness:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Recovery and livelihoods:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim:** Demand: Comment on the resource potential of India's coastline and highlight hazard preparedness. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** Status: Verified cross-cutting route: Comment and highlight · 15 marks · 250 words; this card addresses only tsunami and coastal-preparedness dimensions. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: ITEWC: The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Inundation mapping:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Coastal regulation:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. **Evacuation chain:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Natural buffers:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Ports and critical facilities:** Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. **Fisheries and tourism:** Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. **Community preparedness:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Recovery and livelihoods:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘PYQ DEMAND CARD 2 - 2023 GS-I', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

PYQ DEMAND CARD 3 - 2024 GS-III

Demand: Describe the elements that determine disaster resilience.

Status: Verified direct ownership remains Topic 01; this conservative card routes warning, redundancy, community readiness, natural buffers and livelihood recovery as tsunami-resilience elements.

Model solution: Detection chain: Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **ITEWC:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Local all clear:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Evacuation chain:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Natural buffers:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Community preparedness:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Tsunami Ready boundary:** UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. **Recovery and livelihoods:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. **Warning-outcome firewall:** A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on ‘PYQ DEMAND CARD 3 - 2024 GS-III', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Detection chain: Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **ITEWC:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Local all clear:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Evacuation chain:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Natural buffers:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Community preparedness:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Tsunami Ready boundary:** UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. **Recovery and livelihoods:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. **Warning-outcome firewall:** A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim:** Demand: Describe the elements that determine disaster resilience. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** Status: Verified direct ownership remains Topic 01; this conservative card routes warning, redundancy, community readiness, natural buffers and livelihood recovery as tsunami-resilience elements. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Detection chain: Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **ITEWC:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Local all clear:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Evacuation chain:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Natural buffers:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Community preparedness:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Tsunami Ready boundary:** UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. **Recovery and livelihoods:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. **Warning-outcome firewall:** A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘PYQ DEMAND CARD 3 - 2024 GS-III', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish tsunami generation from tides and cyclone storm surge. Answer in about 150 words.

Model thesis: Claim: Tsunami generation. **Named evidence/example:** A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Tide and storm-surge boundary. **Named evidence/example:** A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Generation is not impact. **Named evidence/example:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Shoaling and inundation. **Named evidence/example:** Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events.
- A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement.
- An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast.
- Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed.

Qualified conclusion: Claim: Tsunami generation. **Named evidence/example:** A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Tide and storm-surge boundary. **Named evidence/example:** A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Generation is not impact. **Named evidence/example:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Shoaling and inundation. **Named evidence/example:** Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Demand decoding: The directive **answer** requires a direct position on 'Distinguish tsunami generation from tides and cyclone storm surge. Answer in about 150 words.', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Tsunami generation. **Named evidence/example:** A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Tide and storm-surge boundary. **Named evidence/example:** A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Generation is not impact. **Named evidence/example:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Shoaling and inundation. **Named evidence/example:** Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Analytical body:

1. **Claim:** A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status,

mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

3. **Claim:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
4. **Claim:** Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Tsunami generation. **Named evidence/example:** A tsunami is a series of long waves generated by sudden displacement of a large water volume, commonly from vertical seabed movement during an undersea earthquake but also from landslides, volcanic activity or other displacement events. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Tide and storm-surge boundary. **Named evidence/example:** A tsunami is not an astronomical tide and not a cyclone storm surge; tides follow gravitational cycles, storm surge is chiefly wind-and-pressure driven, and tsunami generation begins with sudden water displacement. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Generation is not impact. **Named evidence/example:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Shoaling and inundation. **Named evidence/example:** Tsunami waves change as they enter shallower water, producing run-up, inundation and damaging return flow; no universal wave height, speed or inland distance should be assumed. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘Distinguish tsunami generation from tides and cyclone storm surge. Answer in about 150 words.', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain the detection, forecast, warning and local all-clear chain for tsunamis. Answer in about 150 words.

Model thesis: Claim: Detection chain. **Named evidence/example:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** ITEWC. **Named evidence/example:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Service-provider role. **Named evidence/example:** ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Forecast and warning. **Named evidence/example:** A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Local all clear. **Named evidence/example:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment.
- The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities.
- ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority.
- A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact.
- ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities.

Qualified conclusion: Claim: Detection chain. **Named evidence/example:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** ITEWC. **Named evidence/example:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Service-provider role. **Named evidence/example:** ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Forecast and warning. **Named evidence/example:** A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Local all clear. **Named evidence/example:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Demand decoding: The directive **explain** requires a direct position on ‘Explain the detection, forecast, warning and local all-clear chain for tsunamis. Answer in...', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Detection chain. **Named evidence/example:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** ITEWC. **Named evidence/example:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Service-provider role. **Named evidence/example:** ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Forecast and warning. **Named evidence/example:** A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Local all clear. **Named evidence/example:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Analytical body:

1. **Claim:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

3. **Claim:** ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
4. **Claim:** A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
5. **Claim:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Detection chain. **Named evidence/example:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** ITEWC. **Named evidence/example:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Service-provider role. **Named evidence/example:** ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Forecast and warning. **Named evidence/example:** A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Local all clear. **Named evidence/example:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For 'Explain the detection, forecast, warning and local all-clear chain for tsunamis. Answer in...', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Analyse inundation mapping, coastal regulation, evacuation and natural buffers in tsunami preparedness. Answer in about 250 words.

Model thesis: Claim: Inundation mapping. **Named evidence/example:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal regulation. **Named evidence/example:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Evacuation chain. **Named evidence/example:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Natural buffers. **Named evidence/example:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness.
- CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast.
- Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills.
- Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk.

Qualified conclusion: Claim: Inundation mapping. **Named evidence/example:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal regulation. **Named evidence/example:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Evacuation chain. **Named evidence/example:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Natural buffers. **Named evidence/example:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Demand decoding: The directive **analyse** requires a direct position on ‘Analyse inundation mapping, coastal regulation, evacuation and natural buffers in tsunami...', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Inundation mapping. **Named evidence/example:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal regulation. **Named evidence/example:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Evacuation chain. **Named evidence/example:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Natural buffers. **Named evidence/example:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Analytical body:

1. **Claim:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
3. **Claim:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
4. **Claim:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event

owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Inundation mapping. **Named evidence/example:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal regulation. **Named evidence/example:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Evacuation chain. **Named evidence/example:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Natural buffers. **Named evidence/example:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘Analyse inundation mapping, coastal regulation, evacuation and natural buffers in tsunami...', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Examine tsunami preparedness for ports, fisheries, tourism and critical coastal services. Answer in about 250 words.

Model thesis: Claim: Ports and critical facilities. **Named evidence/example:** Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Fisheries and tourism. **Named evidence/example:** Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Community preparedness. **Named evidence/example:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Recovery and livelihoods. **Named evidence/example:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure.
- Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific.
- Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone.
- Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development.

Qualified conclusion: Claim: Ports and critical facilities. **Named evidence/example:** Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Fisheries and tourism. **Named evidence/example:** Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Community preparedness. **Named evidence/example:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Recovery and livelihoods. **Named evidence/example:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Demand decoding: The directive **examine** requires a direct position on ‘Examine tsunami preparedness for ports, fisheries, tourism and critical coastal services....', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Ports and critical facilities. **Named evidence/example:** Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Fisheries and tourism. **Named evidence/example:** Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:**

Community preparedness. **Named evidence/example:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Recovery and livelihoods. **Named evidence/example:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Analytical body:

1. **Claim:** Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
3. **Claim:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
4. **Claim:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Ports and critical facilities. **Named evidence/example:** Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Fisheries and tourism. **Named evidence/example:** Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Community preparedness. **Named evidence/example:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Recovery and livelihoods. **Named evidence/example:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For 'Examine tsunami preparedness for ports, fisheries, tourism and critical coastal services....', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate India's ITEWC-centred tsunami warning architecture while preserving uncertainty and last-mile boundaries. Answer in about 300 words.

Model thesis: Claim: Detection chain. **Named evidence/example:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** ITEWC. **Named evidence/example:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Service-provider role. **Named evidence/example:** ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Forecast and warning. **Named evidence/example:** A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Local all clear. **Named evidence/example:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Community preparedness. **Named evidence/example:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** False alarms and uncertainty. **Named evidence/example:** Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment.
- The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities.
- ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority.
- A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact.
- ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities.
- Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone.
- Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty.
- A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence.

Qualified conclusion: Claim: Detection chain. **Named evidence/example:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** ITEWC. **Named evidence/example:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Service-provider role. **Named evidence/example:** ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Forecast and warning. **Named evidence/example:** A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Local all clear. **Named evidence/example:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Community preparedness. **Named evidence/example:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** False alarms and uncertainty. **Named evidence/example:** Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Demand decoding: The directive **evaluate** requires a direct position on 'Evaluate India's ITEWC-centred tsunami warning architecture while preserving uncertainty and...', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Detection chain. **Named evidence/example:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** ITEWC. **Named evidence/example:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Service-provider role. **Named evidence/example:** ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Forecast and warning. **Named evidence/example:** A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Local all clear. **Named evidence/example:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Community preparedness. **Named evidence/example:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** False alarms and uncertainty. **Named evidence/example:** Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Analytical body:

1. **Claim:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
2. **Claim:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
3. **Claim:** ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
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5. **Claim:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
6. **Claim:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
7. **Claim:** Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
8. **Claim:** A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence ->

competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Detection chain. **Named evidence/example:** Seismic analysis identifies earthquake parameters, while open-ocean and coastal sea-level observations help confirm wave generation and update the assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** ITEWC. **Named evidence/example:** The Indian Tsunami Early Warning Centre at INCOIS integrates earthquake analysis, pre-run model scenarios and sea-level observations to issue tsunami bulletins for Indian and regional service responsibilities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Service-provider role. **Named evidence/example:** ITEWC functions within the UNESCO-IOC Indian Ocean Tsunami Warning and Mitigation System as a Tsunami Service Provider; regional advisory responsibility does not transfer local evacuation authority. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Forecast and warning. **Named evidence/example:** A tsunami assessment may progress from preliminary earthquake information to model-based threat categories and observation-informed updates; each bulletin is evidence at its own stage, not certainty about local impact. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Local all clear. **Named evidence/example:** ITEWC may issue threat-passed or final information, but local conditions can vary and the all-clear decision remains with competent local authorities. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Community preparedness. **Named evidence/example:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** False alarms and uncertainty. **Named evidence/example:** Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For ‘Evaluate India's ITEWC-centred tsunami warning architecture while preserving uncertainty and...', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a sustained coastal-resilience framework for a low-frequency high-consequence tsunami risk. Answer in about 300 words.

Model thesis: Claim: Generation is not impact. **Named evidence/example:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Inundation mapping. **Named evidence/example:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal regulation. **Named evidence/example:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Evacuation chain. **Named evidence/example:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Natural buffers. **Named evidence/example:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Ports and critical facilities. **Named evidence/example:** Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Fisheries and tourism. **Named evidence/example:** Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Community preparedness. **Named evidence/example:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Tsunami Ready boundary. **Named evidence/example:** UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** False alarms and uncertainty. **Named evidence/example:** Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Recovery and livelihoods. **Named evidence/example:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; 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Claim -> named evidence -> analysis -> qualification:

- An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast.
- Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness.
- CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast.
- Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills.
- Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk.
- Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure.
- Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific.
- Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone.
- UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country.
- Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty.
- Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development.
- A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence.

Qualified conclusion: Claim: Generation is not impact. **Named evidence/example:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Inundation mapping. **Named evidence/example:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal regulation. **Named evidence/example:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; 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Demand decoding: The directive **answer** requires a direct position on ‘Design a sustained coastal-resilience framework for a low-frequency high-consequence tsunami...', each clause, risk mechanism, named Indian law/institution/event, dated status, implementation and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Generation is not impact. **Named evidence/example:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Inundation mapping. **Named evidence/example:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal regulation. **Named evidence/example:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; 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3. **Claim:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
4. **Claim:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
5. **Claim:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
6. **Claim:** Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
7. **Claim:** Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
8. **Claim:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Named**

evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

9. **Claim:** UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
10. **Claim:** Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
11. **Claim:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.
12. **Claim:** A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. **Named evidence/example:** Identify the exact Indian law, institution, guideline, warning system, fund, plan or dated event owned by the source. **Analysis:** Connect hazard -> exposure/vulnerability/capacity -> disaster consequence -> competent prevention/response/recovery route. **Qualification:** State source/date/unit/status, mandate, uncertainty, implementation gap, inclusion limit, event/inference boundary or residual risk.

Counter-position / limit: An Act, plan, guideline, scheme, forecast, warning, drill, team, sanctioned capacity, allocation or dispatch cannot alone establish implementation, evacuation, expenditure, reduced loss, equitable recovery or resilience; test mandate, process, status and evidence.

Qualified conclusion: Claim: Generation is not impact. **Named evidence/example:** An undersea earthquake does not automatically create a destructive tsunami; source mechanism, depth, displacement, distance, bathymetry and coastal form influence whether and how waves affect a coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Inundation mapping. **Named evidence/example:** Coastal topography, near-shore bathymetry, source scenarios, land use and exposed assets are combined in inundation or hazard maps for evacuation, siting and preparedness. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Coastal regulation. **Named evidence/example:** CRZ category-specific controls and tsunami hazard mapping support risk-sensitive coastal development; they do not create one universal setback or elevation rule for every coast. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Evacuation chain. **Named evidence/example:** Preparedness requires recognisable warnings, mapped routes, vertical or horizontal refuge options where appropriate, accessible transport, assembly points, accountability and repeated drills. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Natural buffers. **Named evidence/example:** Mangroves, coastal forests, dunes, reefs and other natural features can reduce some impacts and provide co-benefits, but their protection is location-specific and they do not eliminate tsunami risk. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Ports and critical facilities. **Named evidence/example:** Ports, harbours, coastal roads, power, communications, hospitals and fuel or hazardous-material facilities need shutdown, access, continuity and recovery plans tailored to tsunami and multi-hazard exposure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Fisheries and tourism. **Named evidence/example:** Fishers, harbour workers, tourists and seasonal visitors may lack local risk knowledge or face livelihood pressure, so warnings, signage, route information and reopening decisions must be sector-specific. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Community preparedness. **Named evidence/example:** Last-mile safety depends on trusted local relay, natural-warning awareness, inclusive evacuation assistance, drills, family plans and community feedback, not warning-centre capability alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Tsunami Ready boundary. **Named evidence/example:** UNESCO-IOC Tsunami Ready is community-level, indicator-based recognition that requires maintained preparedness; recognition of named communities is not certification of an entire State or country. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** False alarms and uncertainty. **Named evidence/example:** Precautionary action under uncertainty may produce apparent false alarms, but warning evaluation must consider available evidence, missed-event risk, public trust and transparent updating rather than demand impossible certainty. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Recovery and livelihoods. **Named evidence/example:** Recovery should restore ports, fisheries, tourism, housing, services and ecosystems while using safer siting and Build Back Better principles instead of recreating exposed coastal development. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Warning-outcome firewall. **Named evidence/example:** A sensor network, model run, bulletin, siren, map, drill or recognition proves an input or capability; receipt, evacuation, safe shelter, livelihood continuity and avoided loss require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing hazard -> exposure/vulnerability/capacity -> risk/impact -> prevention/mitigation/preparedness -> response/recovery/BBB; write four to seven claim -> named evidence/example -> analysis -> qualification points; reserve the final minute for source, date, unit, status, mandate and event/inference checks.

Why this earns marks: The answer obeys the directive, uses India-centric evidence, and preserves risk terms, legal character, institutional mandate, warning/cycle stage, finance status, implementation and resilience distinctions.

How to improve this answer: For 'Design a sustained coastal-resilience framework for a low-frequency high-consequence tsunami...', replace the weakest generic point with one exact Indian mechanism, institution/guideline/event, dated source and unit, implementation bottleneck, local-capacity or inclusion safeguard and answer-specific qualification.